Running Key Cipher

An encryption method where a long key (usually a book passage which is non-repeating) is used to encrypt a message.

Key : a long, non-repeating key
ex: a book passage.

Algorithm [Encryption]

1. Take plaintext P and running key K
2. Ensure length of key K ≥ length of plaintext P
3. Convert each character of P and K
   Use mapping : A = 0, B = 1, ... Z = 25.
4. For each character i in plaintext :
   $C[i] = (P[i] + K[i]) \bmod 26$
5. Convert numeric values of C[i] back to letters
6. Combine all letters to form ciphertext C

```
BEGIN
  INPUT P, K
  Set C = " "
  For i = 0 TO length(P)-1 Do
     P_VAL = ASCII(P[i]) - ASCI
     K_VAL = ASCII(K[i]) - ASCI
     C_VAL = (P_val + K_val) MOD 26
     C = C + CHAR(c_val + ASCII('A'))
  END FOR
  OUTPUT C
END.
```

Decryption

① Start

② Take Cipher text C and Running key K

⇒ INPUT C, K

SET P = " "

③ For each character position i;

Use row K[i]

Find C[i] in that row

Trace Column → get P[i]

⇒ FOR i = 0 TO Length(C) - 1 DO

C_val = ASCII (C[i]) - ASCII ('A')

K_val = ASCII (K[i]) - ASCII ('A')

P_val = (C_val - K_val + 26) MOD 26

P = P + CHAR (P_val + ASCII ('A'))

END FOR

④ Combine all P[i] → plaintext

OUTPUT P

END

⑤ STOP.